Set
Wed 9 Sep 2026

Problem Set 3 — Functions

Session 3 · Domain, codomain, and undoing

BEFORE YOU START No question here is answered by a verdict alone. "Injective" is not an answer; "injective, because $f(a_1) = f(a_2)$ forces $a_1 = a_2$ since..." is. To *disprove* injectivity or surjectivity you need one concrete witness — to *prove* either, you need an argument covering every element.

Part A · Mechanics

1 IMAGES AND PREIMAGES

Let $f : \mathbb{Z} \to \mathbb{Z}$ be given by $f(n) = 3n - 2$. Compute $f(\{-1, 0, 2\})$, $f^{-1}(\{7\})$ and $f^{-1}(\{5\})$. Explain in one sentence why the third answer is what it is.

2 IMAGE AGAINST CODOMAIN

Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2 - 4x + 7$. State the image of f and prove your answer. Is f surjective? Justify.

3 UNIT CIRCLE

Derive each value from the unit circle. For each, state the quadrant and the reference angle before you give the number. A recalled value without that working is not a derivation.

a. $\cos \frac{2\pi}{3}$
b. $\sin \frac{3\pi}{4}$
c. $\cos \frac{7\pi}{6}$
d. $\sin \frac{5\pi}{3}$
e. $\tan \frac{3\pi}{4}$

Part B · Proof

4 CLASSIFICATION

For each function decide whether it is injective, surjective, both or neither. Prove each positive claim; disprove each negative one with an explicit witness.

a. $f : \mathbb{R} \to \mathbb{R},\ f(x) = 5 - 3x$

b. $f : \mathbb{Z} \to \mathbb{Z},\ f(n) = n^2 - n$

c. $f : \mathbb{R} \to \mathbb{R},\ f(x) = x^3 - 3x$

d. $f : \mathbb{N} \to \mathbb{N},\ f(n) = n^2$

e. $f : \mathbb{R} \setminus \{0\} \to \mathbb{R} \setminus \{0\},\ f(x) = \dfrac{1}{x}$

f. $f : \mathbb{R} \to [-1, 1],\ f(x) = \sin x$

g. $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z},\ f(a, b) = 2a + 4b$

h. $S = \{1, 2, 3\},\ f : \mathcal{P}(S) \to \mathcal{P}(S),\ f(X) = S \setminus X$

5 COMPOSITION

Let $f : A \to B$ and $g : B \to C$.

a. Prove that if $g \circ f$ is surjective, then g is surjective.

b. Show by explicit example that f need *not* be surjective under the same hypothesis.

c. In one sentence, say why (a) is about g and not f, given that in class we proved $g \circ f$ injective forces f injective.

6 RESTRICTION

Take the rule $x \mapsto x^2 - 4x + 7$ from question 2. Choose a domain $A \subseteq \mathbb{R}$ and a codomain B making $f : A \to B$ a **bijection**, and prove both halves.

Part C · Communication

7 FIND THE ERROR

The following was produced by a chatbot. It contains **three** distinct errors. Find all three; for each, say what is wrong and give a witness where one is needed.

> **Claim.** $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \dfrac{x}{x^2+1}$, is a bijection, and its inverse is $f^{-1}(x) = \dfrac{x^2+1}{x}$.
> **"Proof."** f is injective because it is a ratio of polynomials with no repeated factors. It is surjective because its domain and codomain are both $\mathbb{R}$. Inverting a fraction gives the inverse function.

8 REWRITE PRECISELY

Each sentence is sloppy or false as written. Rewrite each so that it is unambiguous and true.

a. "x^2 is not injective."

b. "Every function has an inverse — you just swap x and y."

c. "$\sin^{-1}(x)$ means $\dfrac{1}{\sin x}$."